

Makefile knobs for Verilator UVM

Module twenty-two named Verilator as a plausible UVM host

UVM_HOME and include paths

- Accellera UVM lives outside your DUT tree
- Point UVM home at the library **src** directory that contains ``uvm_pkg.sv``
- This course does not vendor the full Accellera tree
- If UVM home is unset or wrong

SIM, TEST, and UVM test name

- The simulator knob selects the Verilator recipe
- TEST names the SystemVerilog top file
- At run time the binary needs plus UVM test name matching the registered test class
- Read the in-course Makefile

Read the Makefile flow

```
wsl - module23-verilator-uvm-makefile/examples/navigation

# Track A - real Unix

== pwd ==
/mnt/d/proj/designs/learning/courses/learn_uvm2017/examples/verilator-uvm-hello
== tree ==
total 8
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 ..
-rwxrwxrwx 1 yongfu yongfu 2289 Jul 21 10:41 Makefile
-rwxrwxrwx 1 yongfu yongfu 1708 Jul 21 10:46 README.md
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 dut
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 scripts
drwxrwxrwx 1 yongfu yongfu 4096 Jul 21 10:41 tb
dut:
and_gate.v

tb:
test_and_gate_uvm.sv
UVM_HOME unset and archive UVM not found - dry-run will error (expected if Accellera not installed)
```

Real shell — Makefile dry-run

Read the Makefile flow — try these

```
# cd in-course Verilator + UVM hello
cd courses/learn_uvm2017/examples/verilator-uvm-hello

# export UVM_HOME — Accellera UVM 2017 src (uvm_pkg.sv lives here)
export UVM_HOME=/path/to/1800.2-2017-1.0/src

# ls — confirm Makefile, dut/and_gate.v, tb/test_and_gate_uvm.sv
ls

ls dut tb

# make dry-run — verify UVM_HOME and sources without compiling
make dry-run
```

Common fail modes

- Wrong or missing UVM home is the fastest failure
- Running make from the wrong directory misses the Makefile
- Missing Verilator on PATH fails at compile
- Mismatch between top module and plus UVM test name yields no tests executed
- First cold compiles take minutes, that is normal
- Browser sketch success never substitutes for reading these Makefile errors offline

Your turn

- Complete the checklist
- When you are ready